



Case Study 106 : Math Quiz Battle Arena (Competitive Quiz Platform)

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BTECH CSE 25-29

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Problem Statement

- Many online math quiz platforms are static and lack interactive features such as real-time scoring, difficulty levels, competitive ranking, and user discussion systems. Players cannot easily track their improvement or compete with others.
- There is often no structured progression system or time-based challenge to encourage faster problem solving.

Objective

=> The objective of this project is to develop a responsive **Math Quiz web application** using **HTML5, CSS3, and JavaScript** that provides:

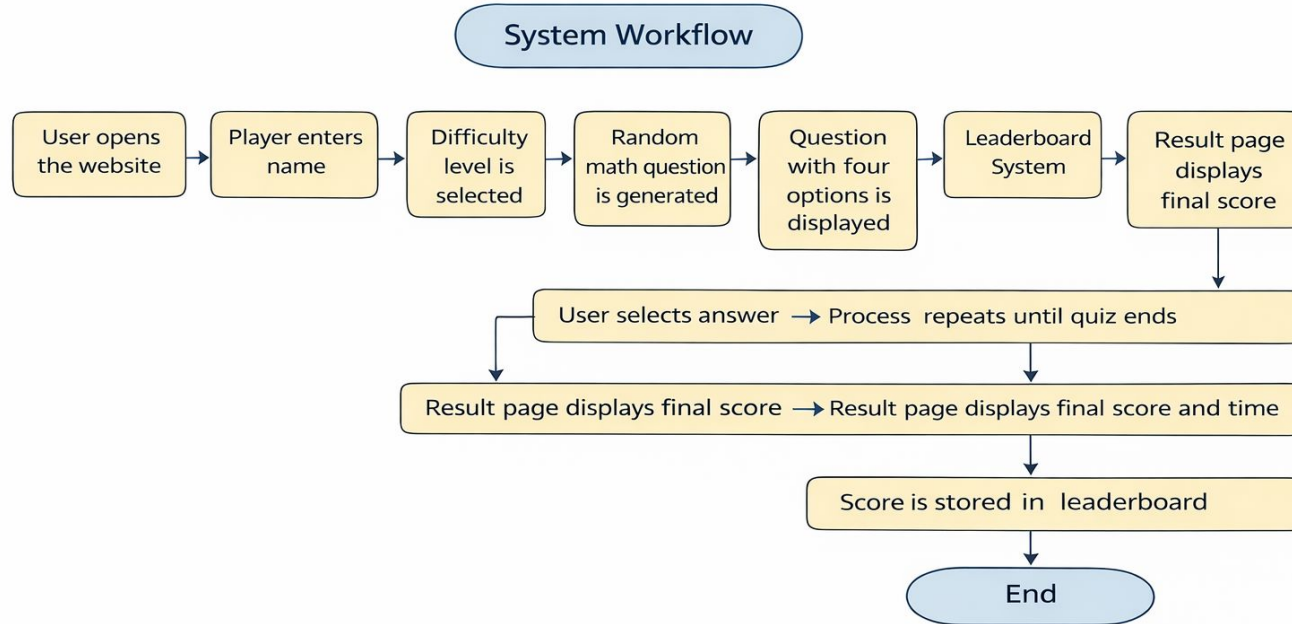
- Timed math quizzes
 - Difficulty levels (Easy, Medium, Hard)
- Real-time score tracking
- Leaderboard ranking system
- Discussion forum for users



Solution

Module / Feature	Implementation	Benefit
Interactive Quiz System	Random math questions generated using JavaScript	Ensures every quiz attempt is unique
Difficulty Levels	Easy, Medium, and Hard levels implemented with different question logic	Allows users to practice according to skill level
Timer Mechanism	10-second countdown timer implemented using setInterval()	Adds challenge and encourages quick problem solving
Real-Time Score Tracking	Score updates immediately when the correct answer is selected	Provides instant feedback to the user
Leaderboard System	Player scores and completion time stored using LocalStorage	Allows users to compete and track progress
Discussion Forum	Comment section implemented with LocalStorage	Enables users to share strategies and interact
Result Evaluation	Result page displays final score and performance message	Helps users evaluate their quiz performance

How it works



How it works

Conclusion

Math Quiz Battle Arena successfully provides an interactive environment for practicing mathematics. The system combines quiz gameplay with competitive ranking and community interaction.

The application demonstrates how modern web technologies can create a complete educational platform without requiring server-side infrastructure.

What It Demonstrates

- Dynamic quiz generation using JavaScript
- Real-time scoring and validation
- Persistent leaderboard using LocalStorage
- Responsive UI design using CSS
- Interactive user engagement through a forum

The project highlights the power of the **frontend** to create engaging educational applications.





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